

CSA15 Cloud Computing and Big Data Analytics

CO4 AT2 - Big Data Industry Problem Solving Guided Tool Workbook

Course Code	CSA1512	Slot	B
Assessment	CO4 AT2	Bloom Level	BL4 - Analyze
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Mode	Individual guided tool-based workbook	Submission	GitHub repository link through Google Classroom
End Evidence	Workbook + screenshots + diagrams + dashboard/mockup	Marks	30

Student Details

Student Name	D.S.Tharuniya Priya	Register Number	192511317
Class / Section	CSA1512	Slot	B
Selected Scenario	Scenario 4 — Cybersecurity Threat Monitoring	Submission Date	19-08-2026
GitHub Link	github.com/tharu-ui/Cloud-Computing-and-BDA	Google Classroom Status	Turned In

Workbook Purpose

What this workbook does

This workbook guides students to use real software tools and official technical references to derive a big-data industry solution. Students should learn from the tool, observe patterns, record evidence and then write their own technical explanation.

- Do not copy full text from websites or AI tools.
- Use official references to justify architecture, processing and security decisions.
- Screenshots are evidence; explanations are the actual assessment answer.

Scenario Bank

Instruction

Faculty may assign one scenario or ask students to complete more than one. If no specific scenario is assigned, select one scenario and complete all tool-guided sections for that scenario.

No.	Scenario	Data Context
1	Healthcare Emergency Analytics	Emergency department data, ambulance updates, bed availability, alerts and treatment delays.
2	E-Commerce Customer Behavior Intelligence	Clickstream events, cart abandonment, payment failure, campaign traffic and recommendation impact.
3	Smart Agriculture Advisory Platform	Weather, soil sensors, crop advisories, pest reports, market prices and farmer app interactions.
4	Cybersecurity Threat Monitoring	Authentication logs, firewall alerts, API access logs, failed login attempts and suspicious network events.

5	Public Transport Demand Analytics	Ticketing data, bus GPS, passenger count, route delay, peak-hour demand and crowding alerts.
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Selected Scenario and Reason

Scenario : Cybersecurity Threat Monitoring Reason: This scenario deals with authentication logs, firewall alerts, API access logs, failed login attempts and suspicious network events a data context that closely matches prior work in digital forensics for my capstone (PulsePay – Cloud based fake payment Screenshot detection).
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Guided Tool Module 1: Use Cloud Architecture Centers

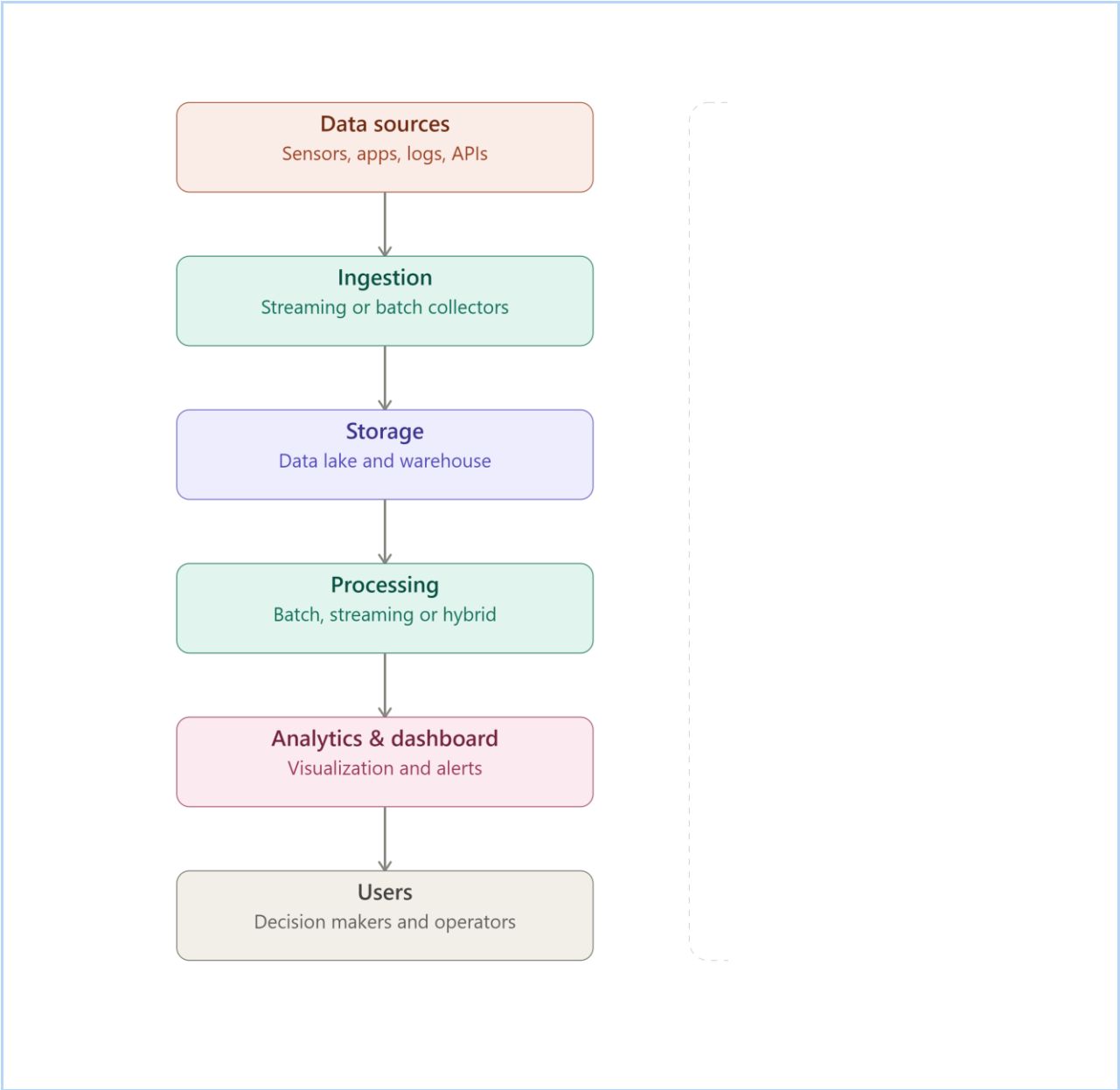
Purpose: Use official cloud architecture references to identify suitable ingestion, storage, processing, analytics and security components for the selected scenario.

No.	Perform	Observe / Decide	Record in Workbook
1	Open AWS Architecture Center, Azure Architecture Center and Google Cloud Architecture Center in three browser tabs.	You are not copying a full architecture. You are learning common cloud patterns and service categories.	Paste the three page titles or URLs.
2	Search each site using terms from your scenario, such as healthcare analytics, streaming analytics, data lake, retail analytics, IoT analytics or security monitoring.	Notice which patterns repeatedly appear: ingestion, data lake, warehouse, processing, ML/analytics, visualization and security.	Record at least one useful reference from each cloud provider.
3	Identify the services/components mentioned in the reference diagrams.	Separate service category from vendor-specific service name. Example: object storage is the category; Amazon S3/Azure Blob Storage/Cloud Storage are service examples.	Complete the tool-category table.
4	Compare the three references.	Choose the most suitable architecture pattern for your scenario and explain why.	Write a short justification.
5	Draw your own provider-neutral architecture in draw.io.	Your diagram must show data sources -> ingestion -> storage -> processing -> analytics/dashboard -> users, with security controls.	Paste the diagram screenshot/link.
Cloud Provider	Reference Page / URL	Useful Pattern Found	Useful Service Categories
AWS	aws.amazon.com/architecture/	real-time log ingestion and threat detection	ingestion, stream processing, data lake, ML anomaly detection, alerting
Azure	learn.microsoft.com/azure/architecture/	SIEM and SOAR reference pattern	log collection, event hub, analytics workspace, dashboards
GCP	docs.cloud.google.com/architecture	Centralized logging pipeline feeding into a security analytics layer	Pub/Sub ingestion, BigQuery, Chronicle/Looker dashboards

Cloud Architecture Justification

The reference architectures across AWS, Azure and Google Cloud follow a consistent pattern for security monitoring: logs and events are first collected through a streaming ingestion service, then stored centrally, processed for anomaly detection, and finally surfaced through a monitoring/alerting layer. For this scenario, a streaming-ingestion-first architecture is most suitable because authentication logs, firewall alerts and failed login attempts must be analyzed close to real time to detect threats before they escalate. A hybrid approach is used for storage, where raw logs are retained in a data lake for historical analysis while a processed, structured layer supports real-time dashboards. This pattern is provider-neutral and can be implemented with any of the three vendors using their equivalent ingestion, storage and analytics services.

Architecture Diagram Evidence



Guided Tool Module 2: Use Hadoop and Spark Documentation

Purpose: Decide whether your problem needs batch processing, streaming processing or hybrid processing by reading official Hadoop/Spark documentation.

No.	Perform	Observe / Decide	Record in Workbook
1	Open the Apache Hadoop MapReduce tutorial.	MapReduce is suitable when large stored data can be processed in batches, such as daily logs or historical transactions.	Write one sentence on what mapper and reducer mean.
2	Open the Apache Spark Structured Streaming guide.	Structured Streaming is suitable when new events continuously arrive, such as sensors, clickstream, live alerts or location updates.	Write one sentence on what streaming source, processing and sink mean.
3	Classify your scenario data into batch data and streaming data.	Historical records normally become batch data; live sensor/log/API updates normally become streaming data.	Complete the batch-vs-stream table.
4	Choose one processing approach for your scenario: Hadoop/MapReduce, Spark batch, Spark streaming or hybrid.	The decision must match the data velocity and decision urgency.	Write your processing decision.
5	Prepare a simple pseudo-flow, not full code.	Example: Mapper reads log line -> emits hour, count. Reducer sums counts. Dashboard shows peak hour.	Write pseudo-flow or paste diagram.
Data Item	Batch / Streaming	Why	Suggested Framework
Authentication logs (login history)	Batch	Historical login records are used for periodic audit and trend analysis, not immediate action	Hadoop MapReduce / Spark batch
Failed login attempts	Streaming	Repeated failed logins indicate a possible brute-force attack in progress and need immediate detection	Spark Structured Streaming
Firewall alerts	Streaming	Firewall events signal active intrusion attempts that require real-time blocking decisions	Spark Structured Streaming
API access logs	Batch	Used mainly for usage pattern analysis and periodic compliance reporting	Hadoop MapReduce / Spark batch
Suspicious network events	Streaming	Anomalous traffic must be flagged as it happens to prevent data exfiltration	Spark Structured Streaming

Processing Framework Decision

This scenario requires a hybrid processing approach. Historical authentication and API access logs are processed in batch mode using Hadoop MapReduce or Spark batch jobs to generate periodic security reports and audit trails. However, failed login attempts, firewall alerts and suspicious network events require Spark Structured Streaming, since these events must be analyzed as they arrive to detect and respond to active threats such as brute-force attacks or intrusion attempts. A hybrid pipeline therefore best matches the mix of data velocity and decision urgency present in this scenario.

MapReduce / Spark Pseudo-Flow

Streaming path: Firewall/login event arrives → Stream processor checks event against threshold rules (e.g., more than 5 failed logins in 2 minutes from same IP) → If threshold breached, mapper emits (IP address, alert type) → Reducer aggregates alert

counts per IP → Dashboard displays real-time alert with risk score.

Batch path: Daily log file is read → Mapper emits (user ID, login timestamp) → Reducer counts logins per user per day → Output stored in warehouse for compliance audit report.

Guided Tool Module 3: Threat Modeling with OWASP Threat Dragon / Microsoft Tool

Purpose: Identify privacy and security risks in the big-data solution before proposing controls.

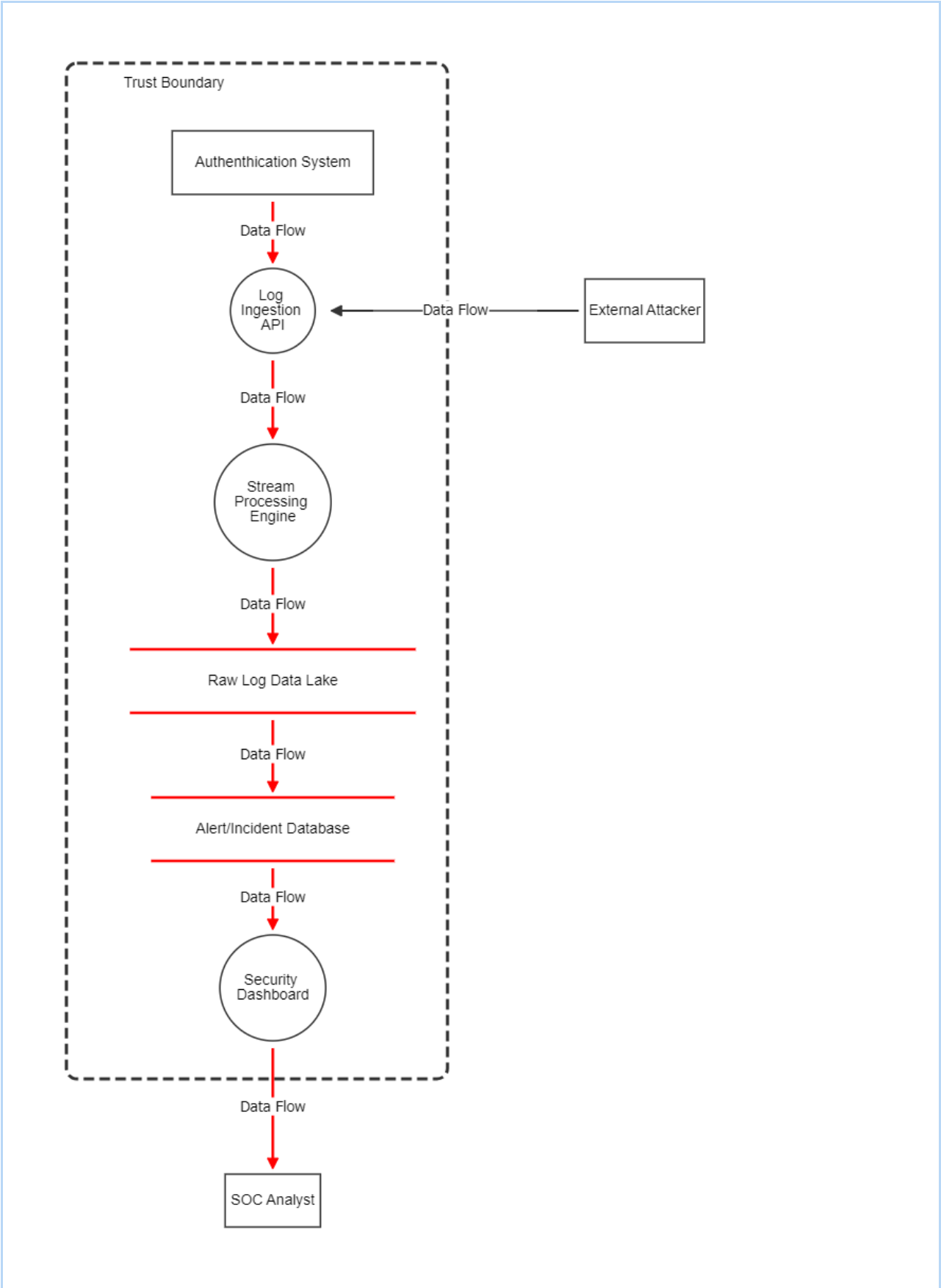
Tool Choice

Use OWASP Threat Dragon if you need a web/desktop cross-platform threat modeling tool. Use Microsoft Threat Modeling Tool when a Windows system is available. The required evidence is the threat model diagram and threat-control table.

No.	Perform	Observe / Decide	Record in Workbook	
1	Open OWASP Threat Dragon or Microsoft Threat Modeling Tool.	A threat model is a security diagram that shows components, data flows, trust boundaries and possible threats.	Write which tool you used.	
2	Create a new threat model named with your scenario.	Use a clear title such as CO4_AT2_Healthcare_Emergency_Analytics.	Paste model name.	
3	Add external entities.	External entities are users/systems outside your platform: patient, customer, sensor, payment gateway, bus GPS device, public API.	List at least two external entities.	
4	Add processes and data stores.	Processes transform data; data stores hold data. Example: ingestion API, stream processor, data lake, warehouse, dashboard.	List processes and stores.	
5	Draw data flows and trust boundaries.	Trust boundaries mark where security assumptions change: public internet to cloud, app to API, processing layer to storage.	Paste screenshot of DFD.	
6	Identify threats using STRIDE thinking.	Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service and Elevation of Privilege help structure risks.	Complete the threat table.	
7	Write mitigations.	Controls may include authentication, authorization, encryption, masking, audit logs, rate limits, backup and monitoring.	Map every threat to one control.	
Component / Data Flow		Threat	Impact	Mitigation
Log Ingestion API		Spoofing – attacker sends forged log entries pretending to be a legitimate system	False alerts or masking of real attacks	API key/token-based authentication for all log sources
Login event data flow		Tampering – log data altered in transit before reaching storage	Corrupted evidence, missed real threats	TLS encryption in transit + checksums/hash validation
Alert/Incident Database		Repudiation – attacker denies performing malicious login attempts	Inability to prove incident occurred	Immutable audit logging with timestamps
Raw Log Data Lake		Information Disclosure – unauthorized access exposes sensitive log data (IPs, usernames)	Privacy breach, compliance violation	Role-based access control + encryption at rest
Stream Processing		Denial of Service –	Delayed real threat	Rate limiting + auto-

Engine	attacker floods system with fake events to overwhelm processing	detection	scaling on ingestion layer
Security Dashboard	Elevation of Privilege – low-privilege user gains admin access to dashboard	Unauthorized control over alert handling	Strict RBAC + multi-factor authentication for analysts

Threat Model Diagram Evidence



Guided Tool Module 4: Dashboard or Analytics Output Mockup

Purpose: Show how raw big data becomes useful decisions through a dashboard, alert, forecast, recommendation or report.

Beginner-Friendly Choice

Use any one: Power BI, Looker Studio, Tableau Public or Canva. If login/install is difficult, Canva dashboard mockup is acceptable. If data connection is possible, Power BI/Looker Studio/Tableau Public is stronger.

No.	Perform	Observe / Decide	Record in Workbook
1	Prepare a small sample dataset in Excel or Google Sheets with 10-20 rows.	Columns should match your scenario. Example: timestamp, user/location, event type, value, status, risk level.	Paste column names.
2	Choose the dashboard tool.	Power BI/Tableau/Looker are analytics tools; Canva is acceptable for visual mockup when live data connection is blocked.	Write chosen tool and reason.
3	Import or recreate the sample data.	In Power BI/Looker/Tableau, connect data. In Canva, manually create charts from sample values.	Paste screenshot of data/chart setup.
4	Create at least three visual elements.	Use KPI card, bar chart, line chart, table, map, alert count or pie/donut chart as appropriate.	List the visuals.
5	Add decision labels.	A dashboard is useful only if it supports a decision: allocate beds, detect fraud, reroute buses, restock products, warn farmers.	Write decision supported by each visual.
6	Export/share screenshot.	Paste the final dashboard screenshot or public/report link if available.	Paste evidence.

Dataset Column	Meaning	Example Value
Timestamp	Time the security event occurred	2026-08-19 14:32:07
Source IP	Origin address of the login/API attempt	192.168.14.22
Event Type	Category of security event	Failed Login / Firewall Block / API Access
User/Account	Account associated with the event	analyst_23
Status	Outcome of the event	Success / Blocked / Flagged
Risk Level	Severity assigned to the event	Low / Medium / High

Visual Element	Metric Shown	Decision Supported
KPI Card	Total failed login attempts (last 24h)	Flag accounts for temporary lockout
Bar Chart	Event count by risk level (Low/Medium/High)	Prioritize which alerts the SOC team reviews first
Line Chart	Suspicious events over time (hourly)	Identify attack time windows and staffing needs
Table	Top 5 source IPs by blocked attempts	Decide which IPs to add to firewall blocklist

Dashboard / Analytics Output Evidence

The dashboard mockup translates raw security log data into decision-oriented visuals. The KPI card highlighting failed login attempts allows the SOC team to quickly assess ongoing brute-force activity, while the bar chart of events by risk level supports triage prioritization. The line chart of suspicious events over time helps identify peak attack windows for resource planning, and the top-IP table directly supports the decision to update firewall block rules. Together, these visuals convert raw log volume into actionable security response decisions.

Final CO4 AT2 Solution Sections

Final Assembly

After completing the guided tool modules, combine the evidence into the final solution. The final answer must show technical reasoning, not only screenshots.

1. Problem Understanding

What to include: Identify stakeholders, decisions required, data produced, expected analytics output and urgency.

Student Response

Security operations teams face the challenge of monitoring continuous streams of authentication logs, firewall alerts, API access records and network events to detect malicious activity before it causes damage. The primary stakeholders are SOC analysts, IT security administrators and end users whose accounts may be compromised. Key decisions required include identifying brute-force login attempts, blocking suspicious IP addresses, and prioritizing which alerts require immediate investigation. The data produced is high-velocity and continuous, and the expected analytics output is a real-time dashboard with risk-scored alerts. Given that active attacks can escalate within minutes, the urgency of detection and response is high.

2. 5Vs Mapping

What to include: Map Volume, Velocity, Variety, Veracity and Value using scenario evidence.

Student Response

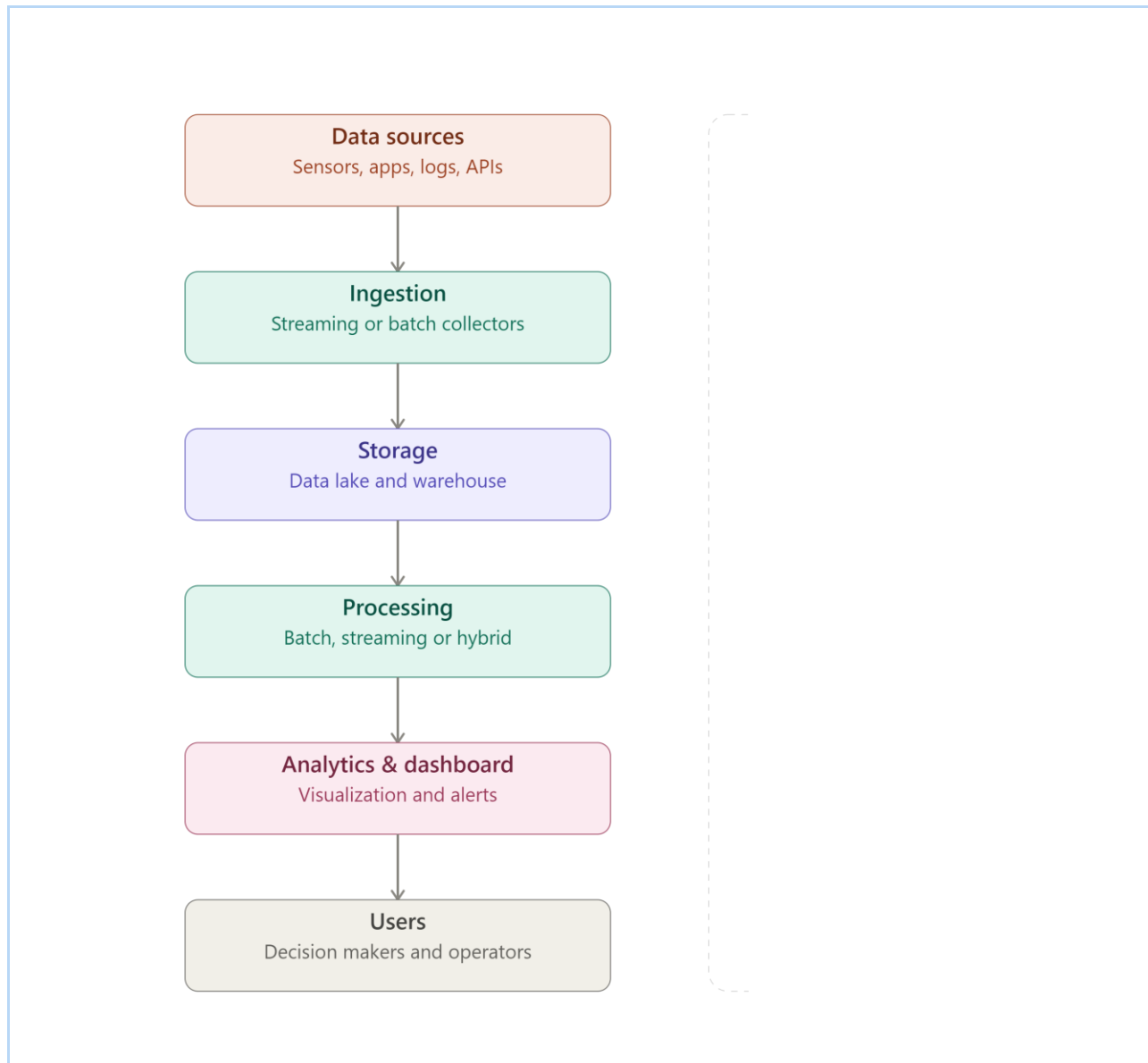
Volume: The platform generates large volumes of log data daily from authentication systems, firewalls and APIs across the organization. Velocity: Failed logins, firewall alerts and network anomalies arrive continuously and must be processed in near real time. Variety: Data includes structured logs (timestamps, IPs, status codes), semi-structured API logs and alert metadata. Veracity: Log data can include noise such as false positives from misconfigured systems, requiring validation before triggering alerts. Value: The output enables faster threat detection, reduced response time to incidents, and stronger compliance with security audit requirements.

3. Big Data Architecture

What to include: Paste final draw.io architecture showing sources, ingestion, storage, processing, analytics/security and users.

Student Response

The architecture follows a streaming-first pattern since most critical events require near real-time detection. Data flows from distributed log sources through a centralized ingestion layer, is stored in both raw and processed forms, and is analyzed continuously before reaching the security dashboard used by analysts.



4. Data Flow Pipeline

What to include: Explain how raw data moves, gets cleaned, stored, processed and converted into dashboard/alert/decision.

Student Response

Raw log data is collected from authentication systems, firewalls and API gateways through a streaming ingestion service. The data is then validated and cleaned to remove malformed or duplicate entries before being written to a data lake for long-term storage. A stream processing engine simultaneously applies detection rules, such as threshold-based checks for repeated failed logins, to identify anomalies. Detected anomalies are converted into structured alerts and stored in an alert/incident database. Finally, this processed data is presented on a security dashboard, where SOC analysts review risk-scored alerts and take response decisions such as blocking IPs or escalating incidents.

5. Tool and Technology Justification

What to include: Use cloud architecture references and Hadoop/Spark documentation to justify chosen tools.

Student Response

Cloud architecture references from AWS, Azure and Google Cloud consistently recommend a streaming-ingestion-to-analytics pattern for security monitoring use cases, which was adopted for this scenario. Based on Apache Spark Structured Streaming

documentation, streaming processing was selected for time-sensitive events such as failed logins and firewall alerts, since Spark supports continuous processing with low latency. Hadoop MapReduce batch processing was retained for historical log analysis and compliance reporting, where immediate response is not required. This combination ensures the platform can handle both real-time threat detection and periodic security auditing.

6. Security and Privacy Strategy

What to include: Use threat model output to explain threats and controls.

Student Response

The threat model developed using OWASP Threat Dragon identified risks across the ingestion, processing, storage and presentation layers, including spoofing of log sources, tampering with data in transit, denial-of-service against the processing engine, and unauthorized access to stored logs. Corresponding mitigations include token-based authentication for all log sources, TLS encryption for data in transit, rate limiting on the ingestion layer, and role-based access control combined with encryption at rest for stored data. Additionally, immutable audit logging was proposed to prevent repudiation of security-relevant actions, and multi-factor authentication was recommended to prevent unauthorized privilege escalation on the analyst dashboard.

7. Challenges and Solutions

What to include: Write at least four big-data implementation challenges and one practical solution for each.

Student Response

First, the high velocity of incoming log data can overwhelm the processing pipeline during peak attack periods; this is addressed through auto-scaling of the stream processing layer and rate limiting at ingestion. Second, distinguishing genuine threats from false positives is difficult due to noisy log data; this is addressed by applying threshold-based and pattern-based detection rules rather than flagging every anomaly. Third, storing sensitive authentication data raises privacy concerns; this is addressed through encryption at rest and strict role-based access control. Fourth, coordinating real-time alerts with analyst response time can create delays; this is addressed by risk-scoring alerts so that only high-severity incidents trigger immediate notifications, reducing alert fatigue.

8. Business Outcome

What to include: State measurable outcome: faster decision, reduced fraud, lower waiting time, improved prediction, lower downtime or better user experience.

Student Response

Implementing this big data security monitoring solution is expected to reduce the average time to detect and respond to security incidents, since threats are flagged in near real time rather than discovered during periodic manual log review. This directly lowers the risk of successful brute-force attacks and unauthorized data access, while also improving compliance reporting through consistent audit trails. Overall, the solution supports faster, evidence-based security decisions and reduces the operational burden on the SOC team.

Rubric

Criteria	Marks	Evaluation Focus	Evidence
Problem and 5Vs analysis	6	Scenario-specific analysis and correct big-data reasoning	Workbook tables
Architecture and data flow	6	Correct pipeline, cloud components and flow clarity	draw.io diagram
Processing framework decision	5	Batch/streaming/hybrid justification using Hadoop/Spark concepts	Decision table
Security and privacy	5	Threat model, risks and controls	Threat model table/screenshot
Dashboard/output design	4	Useful decision-oriented analytics output	Dashboard/mockup
Clarity and submission quality	4	Organized, original, complete evidence	GitHub/GCR

Submission Checklist

- Student details and selected scenario are filled.
- Cloud architecture references are recorded.
- Hadoop/Spark processing decision is completed.
- Threat model diagram/table is included.
- Dashboard/mockup screenshot or link is included.
- Final solution sections are completed.
- DOCX/PDF and supporting files are uploaded to GitHub.
- GitHub repository link is submitted through Google Classroom.

Official Reference Links

AWS Architecture Center: <https://aws.amazon.com/architecture/>

Azure Architecture Center: <https://learn.microsoft.com/en-us/azure/architecture/>

Google Cloud Architecture Center: <https://docs.cloud.google.com/architecture>

Apache Hadoop MapReduce Tutorial: <https://hadoop.apache.org/docs/stable/hadoop-mapreduce-client/hadoop-mapreduce-client-core/MapReduceTutorial.html>

Apache Spark Structured Streaming: <https://spark.apache.org/docs/latest/streaming/index.html>

Microsoft Threat Modeling Tool releases: <https://learn.microsoft.com/en-us/azure/security/develop/threat-modeling-tool-releases>

OWASP Threat Dragon: <https://owasp.org/www-project-threat-dragon/>

Power BI documentation: <https://learn.microsoft.com/en-us/power-bi/>

Looker Studio create report: <https://docs.cloud.google.com/data-studio/create-a-report>

Tableau Public: <https://www.tableau.com/community/public>

Canva Charts: <https://www.canva.com/help/charts/>